

Fertilizer Recommendation and Crop Suggestion System Using Machine Learning

Abstract

Agriculture has a vital part in keeping food security stable and also helping economic growth, especially in developing countries where farming is basically still the main livelihood. In real life, farmers will find themselves in a dilemma when they have to select appropriate fertilizers and crops, as the soil conditions can vary considerably and the environment is also evolving, such as temperature and humidity and more. The use of fertilizer can diminish the yield, accelerate the degradation of soil and harm the wider environment if used incorrectly. The proposed setup in this work may be used for the recommendation of fertilizer using a machine learning approach, and a suggestion workflow of crops for farmers to rely on more factual decisions. The method is based on the soil nutrient properties, environmental factors, and other crop related attributes, and then forecasts the most appropriate fertilizer and crops. Several machine learning models are tested such as Random Forest, Gradient Boosting, XGBoost, and Support Vector Machine (SVM). Based on the experiments it would appear that ensemble learning is superior to the more traditional approaches, and that XGBoost outperforms all the other models. Overall, the proposed system supports sustainable agriculture by making fertilizer management smarter, raising crop productivity, and enabling a more efficient use of resources.

Keywords : machine learning , agriculture, fertilizer recommendation , crop prediction , xgboost , random forest, precision agriculture .

I. INTRODUCTION

One of the most important sectors contributing to the economic growth and food security of the world is agriculture. Properly using fertilizers and the climatic factors are also important to crop productivity apart from soil fertility. It is the farmers who generally find it difficult to select the proper fertilizer and crop, as they have to cope with the changing nature of the soil and the surrounding environment and the vagaries of the seasons as well. If not applied properly, fertilizers can result in reduced crop production, soil deterioration, increased production costs and negative consequences on the environment—Yup, more than anyone would expect.

Traditional agricultural practice relies heavily on the experience and universal advice of experts and it doesn't work for every farm. In recent years, technologies powered by data are rising at a very high speed and machine learning has emerged as a very promising technology for agricultural decision support. Specifically, learning algorithms can decipher complex relationships between soil nutrients, weather conditions and crop requirements to more accurately recommend fertilizers and other crops.

In this work, a machine learning based fertilizer recommendation ,and crop suggestion setup is proposed. It is based on soil related parameters such as Nitrogen, Phosphorus, Potassium (NPK), soil pH, soil moisture, soil organic carbon, among other environmental parameters including temperature, humidity, rainfall, season and irrigation type. Several machine learning models (Random Forest, Gradient Boosting, XGBoost, and Support Vector Machine (SVM)) have been trained, and they are also checked and compared for their performance. Overall, the idea is to raise crop productivity a bit more, cut down on over applying fertilizers and back up sustainable farming habits by using smarter, more data driven recommendations.

II. LITERATURE REVIEW

Abdullah shares a study that demonstrates what you can predict about soil fertility from the pH, nitrogen, phosphorus and potassium and moisture data. Normalization is conducted along with a bit of feature selection, indirectly. Then, the models of Decision Tree model, SVM model and even Random Forest are tested, evaluated for their results. The very toughest one, Random Forest, is less prone to overfitting even in the presence of some noise makes it the best! [1]

Kumar and Patel kind of suggest a soil fertility analysis, and crop prediction based on the soil nutrient and environmental information. They are utilizing Logistic Regression and a Decision Tree model, and for some reason the Decision Tree out performs in predicting fertility classes, not to mention when it comes to recommending crops. [2]

The review in Expert Systems with Applications kind of looks at machine learning methods for predicting soil fertility, and yeah it basically goes on to say that, ensemble along with hybrid approaches tend to beat the single, standalone algorithms. it also stresses that preprocessing and feature engineering are not just “nice to have” but pretty crucial, like in a very practical way. [3]

Sharma et al. create a model for prediction of crop yield by including soil and environmental parameters, it sort of links those together. Based on the results obtained, it can be seen that ensemble models give noticeable results compared to more traditional models which they compared. [4]

Kammerlander et al. combine satellite data, weather data and soil parameters for prediction. The study appears to indicate that blending multi source data together makes it more accurate, quite a bit more accurate anyhow. [5]

A real time soil fertility and crop prediction system is based on data from sensors, machine learning algorithms and provides a reliable soil fertility view and more or less accurate crop suggestions. [6]

Singh and Verma have also used a Decision Tree to classify soil fertility, and Naïve Bayes. Actually, the Decision Tree looks like a better all round solution, although not strictly speaking linear. [7]

Reddy and Kumar are developing a crop recommendation system, similar to this, using soil and climate parameters. In their experiments Random Forest gives better accuracy than the SVM approach for crop selection tasks. [8]

Then there is Ali and Noor smart agriculture model which predicts suitable crops and also fertilizers by applying the machine learning, it exhibits how effective it is in real world, practical application and at the same time it shows its scalability. [9]

Kinda compare Decision Tree with SVM and also Random Forest models Sharma and Gupta note that ensemble methods appear to be more successful than traditional models in the agri context. [10]

Deep learning framework to predict crops, it's more or less about how to do better with neural networks, Ahmed and Rahman introduce it. The neural network models work better than the conventional machine learning models as it can identify more complex relationships between the features, which is relevant in crop forecasting. [11]

For soil fertility predicting, a study has been conducted which included models like Random Forest, Decision Tree, SVM and KNN. The accuracy of the Decision Tree appears to improve in most cases and the Random Forest appears to be more consistent, such as fewer fluctuations in results. [12]

They suggest a system for soil fertility prediction, and a crop suggestion using Random Forest, and they compare this to a Decision Tree, and end up with a more accurate prediction system and more “appropriate” soil fertility suggestions. [13]

In a study comparing soil assessment and crop fertility forecast, SVM, Random Forest and Artificial Neural Networks are put to the test. In this setup the neural networks end up performing the best, mostly because they are able to model tricky, non-linear relations, like there is more going on under the hood. [14]

They did soil fertility estimation study, which employed a combination of XGBoost and neural nets, not just one of them. In practice XGBoost usually achieves higher accuracy, but the neural network bit is responsible for generalisation so it's not just memorising the training data and bouncing better at new data. [15]

In this paper, Jeong et al. use climate and soil data for predicting the yield of crops, in addition to using Random Forest and Artificial Neural Networks. Generally, Random Forest provides more consistent predictions and neural networks capture the nonlinear patterns fairly well. [16]

A precision agriculture system using sensor signals and soil information and weather type environmental information was assembled by Pantazi et al. They employ SVM and RF to classify the condition of the crops fairly well, which will support them in taking timely farming decisions and action planning. [17]

III. METHODOLOGY

A. Task Description

To, of course, create an intelligent fertilizer suggestion system, similar to the machine learning methods. The concept is to analyse soil nutrient content, environmental (surroundings) conditions and also crop associated data, in order to identify the best available fertilizer to enhance crop productivity and fertilize soil. The agricultural signals it uses as input features are Soil Type, Soil pH, Soil Moisture, Organic Carbon, Electrical Conductivity, Nitrogen, Phosphorus, Potassium levels, Temperature, Humidity, Rainfall, Crop Type, Crop Growth Stage, Season, Irrigation Type, Previous Crop, Region, Fertilizer Used Last Season and Yield Last Season.

All that data is then subjected to some preprocessing such as data cleaning, handling missing values, encoding categorical features, scaling features and the additional feature engineering. We also create a number of more derived features like NP Ratio, PK Ratio, NK Ratio, Total NPK, Temperature-Humidity Interaction, Moisture-Soil Type Interaction, and NPK per pH, primarily to maintain the predictive performance. After that, a feature selection mechanism, meaning based on the Chi-Square statistical test is used to figure out which features are the most relevant for fertilizer recommendation.

The chosen features are also subsequently used to train some of the machine learning techniques such as Random Forest, Gradient Boosting, XGBoost and Support Vector Machine (SVM) machines as well. Hence, which of the approaches is the most efficient for prediction using metrics like the Accuracy, Precision, Recall and the F1-Score is evaluated afterwards. In general, the system can provide a robust

data-driven fertilizer recommendation system and help boost farm productivity while also enabling more sustainable farming practices for the farmers.

B. Dataset Description

The information in the proposed Fertilizer Recommendation and Crop Prediction System includes soil characteristics, environmental factors, crop characteristics and cropping practices. In addition, it also contains important parameters such as Soil Type, Soil pH, Soil Moisture, Organic Carbon, Electrical Conductivity, and nutrients like Nitrogen, Phosphorus and Potassium levels. There are other weather related values, namely, Temperature, Humidity, Rainfall and Crop Type and Season. It records Irrigation Type, Previous Crop on the farm and even Yield Last Season for farm management.

The dataset contains both numerical as well as categorical fields, which helps with assessing soil fertility and crop needs. The data underwent several pre-processing steps, such as data cleaning, categorical encoding, feature scaling, and some feature engineering, even prior to actual model training, to improve the overall data quality. This should also help with prediction accuracy in some way as well, albeit an indirect way for now.

The variable to be predicted in the data is Recommended_Fertilizer which is basically the best fertilizer for the soil condition and environment, more or less. The processed data is then provided to the train and evaluation pipelines of machine learning models, such as Random Forest and Support Vector Machine (SVM), Gradient Boosting and KNN to provide more accurate fertilizer recommendations and also crop prediction, all in one go.

TABLE I: Description of the data sets and attributes of the features

Attribute	Count
Total Records	19,689
Total Features	16
Crop Types	55
States	30
Seasons	6
Year Range	1997–2020

The dataset includes a wide variety of agriculture data that represents the soil characteristics, climatic conditions and cultivation methods of different areas, thus, establishing a sound basis for building an accurate fertilizer recommendation system.

IV. MODEL ARCHITECTURES

A. Feature Selection using Chi-Square Test

If the number of features is large, apply filter feature selection algorithm: A.1 Feature Selection using Chi-Square Test.

The target variable of the data set is Recommended_fertilizer, which is the best fertilizer for the particular soil and climatic conditions, more or less. The processed data is then used to train and test the machine learning models such as Random Forest, Support Vector Machine (SVM), Gradient Boosting and KNN to provide more accurate recommendations for fertilizer dosages and also crop prediction, kinda in one stroke.

The Chi-Square statistic is given by:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

where O_i represents the observed frequency and E_i represents the expected frequency.

Initially attempts were made to develop some more additional attributes such as NP Ratio, PK Ratio, NK Ratio, Total NPK and Temperature–Humidity Interaction through feature engineering techniques. Even Moisture–Soil Type Interaction came into play, along with NPK per pH in the mix. Later SelectKBest was applied in order to be able to select the highest ranked features. The selected features were then fed into the machine learning models, in essentially the above order. Such a full arrangement can help improve the classification power and reduce computational effort.

A. Random Forest (RF)

Random Forest (RF) is a bit like an ensemble or bagging machine learning method where it trains a forest of decision trees, the trees are trained on bootstrapped data and there is also an element of feature selection. The final prediction, or end result, is the result of voting by the majority of these trees, and you typically will see higher classification accuracy and less overfitting. It is somewhat versatile as far as working with high-dimensional agriculture data and nonlinear patterns, and it therefore becomes somewhat well suited for fertilizer recommendation systems .

Random Forest prediction function is:

$$\hat{Y} = \text{Mode}\{T_1(X), T_2(X), \dots, T_n(X)\}$$

where $T_i(X)$ represents the prediction of the i^{th} decision tree and $\hat{Y}$ denotes the predicted fertilizer class.

For this reason, we used two different configurations to train the Random Forest classifier: using the full feature set and using the set selected via Chi-Square. To test this, we used two training setups of the Random Forest classifier: with all features and with the features selected by Chi-Square.

A.1 Random Forest without Chi-Square Feature Selection

The baseline Random Forest model got an overall classification accuracy of 88% along with a weighted F1-score of 0.87 . The classifier's F1 scores of 0.94, 0.88 and 0.95 for fertilizer Classes 1, 2 and 5, respectively, seem to be quite good. However, Class 4 had relatively poor results primarily due to the small number of training samples. Relying on the simpler data that the classifier could get to make the more complex links between soil nutrients, soil conditions, and fertilizer classes however, the fertiliser recommendations made were broadly accurate.

Table II. Classification Report of Random Forest (Without Chi-Square)

Class	Precisi on	Reca ll	F1-Sco re	Suppo rt
Compost	0.79	0.83	0.81	199
DAP	0.96	0.93	0.94	584
MOP	0.91	0.86	0.88	282
NPK	0.61	0.77	0.68	128
SSP	0.17	0.05	0.08	37
Urea	0.95	0.94	0.95	620
Zinc Sulphate	0.68	0.79	0.73	150
Accuracy	–	–	0.88	2000
Macro Avg	0.72	0.74	0.72	2000
Weighted Avg	0.88	0.88	0.87	2000

A.2 Random Forest with Chi-Square Feature Selection

The overall classification accuracy was maintained at approximately 88% after applying Chi-Square feature selection and the weighted F1-score actually increased from 0.87 to 0.88 with the Random Forest classifier. The precision and recall results of some fertilizer classes showed only marginal improvements, indicating that the elimination of the redundant features did not help the consistency of predicting the fertilizer class and did not badly affect the accuracy of predictions. To put it differently, these results indicate that Random Forest produces fairly stable results with a smaller number of features.

Table III. Classification Report of Random Forest (With Chi-Square)

Class	Precisi on	Reca ll	F1-Sco re	Suppo rt
Compost	0.79	0.83	0.81	199
DAP	0.96	0.93	0.94	584
MOP	0.91	0.86	0.88	282

NPK	0.61	0.77	0.68	128
SSP	0.17	0.05	0.08	37
Urea	0.95	0.94	0.95	620
Zinc Sulphate	0.68	0.79	0.73	150
Accuracy	–	–	0.88	2000
Macro Avg	0.72	0.74	0.72	2000
Weighted Avg	0.88	0.88	0.87	2000

B. Gradient Boosting (GB)

Gradient Boosting is just an ensemble learning method in which a collection of weak learners are combined, but not all in one. They are combined in order to minimize the prediction errors and therewith to enhance classification performance. Each time, a new decision tree is learnt kinda to correct the errors of the previous trees, and at the end a good predictive model will be produced.

The prediction model is written as:

$$F(x) = \sum_{m=1}^M \gamma_m h_m(x)$$

where $h_m(x)$ is the weak learner and γ_m represents its corresponding weight.

The complete feature set was fed into the classifier, along with the features picked up by the Chi-Square, kind of as a test to see what happens if I test the full, and the features picked by Chi-Square.

B.1 Gradient Boosting without Chi-Square Feature Selection

The overall classification accuracy of the Gradient Boosting model (baseline model) was approximately 87% and the weighted F1-score was 0.87. The classifier was fairly consistent with the major fertilizer classes and provided reasonably balanced precision and recall numbers – not too far off. But for the minority categories, the prediction accuracy appeared a little less, possibly due to the lack of balance in the classes, yeah.

Table IV. Classification Report of Gradient Boosting (Without Chi-Square)

Class	Precisi on	Reca ll	F1-Sco re	Suppo rt
Compost	0.91	0.79	0.85	199
DAP	0.94	0.93	0.94	584
MOP	0.89	0.84	0.87	282
NPK	0.78	0.71	0.75	128

SSP	0.10	0.11	0.11	37
Urea	0.90	0.95	0.92	620
Zinc Sulphate	0.71	0.78	0.75	150
Accuracy	–	–	0.87	2000
Macro Avg	0.75	0.73	0.74	2000
Weighted Avg	0.87	0.87	0.87	2000

B.2 Gradient Boosting with Chi-Square Feature Selection

After Chi-Square feature selection, the Gradient Boosting classifier kept a total classification accuracy around 87% still. Even though the overall accuracy didn't really budge, the macro-average F1 score went up from 0.74 to 0.75, which suggests less trouble with per-class predictions. Put another way, the selected features were somehow able to reduce redundant signals but retained the ability to predict the model's power.

Table V. Classification Report of Gradient Boosting (With Chi-Square)

Class	Precision	Recall	F1-Score	Support
Compost	0.92	0.79	0.85	199
DAP	0.97	0.93	0.95	584
MOP	0.87	0.84	0.86	282
NPK	0.78	0.71	0.75	128
SSP	0.15	0.19	0.17	37
Urea	0.88	0.95	0.92	620
Zinc Sulphate	0.71	0.79	0.75	150
Accuracy	–	–	0.87	2000
Macro Avg	0.76	0.74	0.75	2000
Weighted Avg	0.88	0.87	0.87	2000

C. K-Nearest Neighbors (KNN)

K-Nearest Neighbors (KNN) is a supervised learning method, which basically sorts a new sample by looking at the most common class that shows up among its closest neighbor instances. Also, KNN requires distance calculations, which can cause the outcome to fluctuate greatly if there are irrelevant features, or even features that are similar to one another such as duplicates.

The Euclidean distance is given by the following formula:

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

The classifier got evaluated on both the original feature set, and the Chi-Square chosen features set a bit separately.

C.1 KNN without Chi-Square Feature Selection

The F1 score of the baseline KNN model was around 0.42 with an overall classification accuracy of around 45% which is close. In practice the classifier's performance concerning the precision was found to be less than the one of the original classifiers and also concerning the recall for most of the fertilizer classes, since these redundant features somehow confused the distance based decision making. So these results show that KNN is sensitive to high-dimensional feature spaces but, guess what, it does not like extra dimensions when all other factors seem to be OK.

Table VI. Classification Report of KNN (Without Chi-Square)

Class	Preci sion	Reca ll	F1-Sco re	Suppo rt
Compost	0.27	0.21	0.23	199
DAP	0.41	0.56	0.47	584
MOP	0.33	0.20	0.25	282
NPK	0.18	0.09	0.12	128
SSP	0.00	0.00	0.00	37
Urea	0.61	0.73	0.66	620
Zinc Sulphate	0.16	0.07	0.10	150
Accuracy	—	—	0.45	2000
Macro Avg	0.28	0.27	0.26	2000
Weighted Avg	0.40	0.45	0.42	2000

C.2 KNN with Chi-Square Feature Selection

The baseline KNN model reached overall classification accuracy 45% with a weighted F1 score around 0.42. It looked like the classifier had lower precision and recall for most fertilizer classes , because redundant features were still there and kinda messed up the distance based decision making. Overall these outcomes suggest that KNN is quite sensitive to high dimensional feature spaces, even if the signal is kinda there.

Table VII. Classification Report of KNN (With Chi-Square)

Class	Precisi on	Reca ll	F1-Sco re	Suppo rt
Compost	0.66	0.72	0.69	199
DAP	0.88	0.86	0.87	584
MOP	0.80	0.77	0.78	282
NPK	0.61	0.57	0.59	128
SSP	0.13	0.05	0.08	37
Urea	0.86	0.90	0.88	620
Zinc Sulphate	0.60	0.62	0.61	150
Accuracy	–	–	0.80	2000
Macro Avg	0.65	0.64	0.64	2000
Weighted Avg	0.79	0.80	0.79	2000

D. Support Vector Machine (SVM)

Support Vector Machine (SVM) is a supervised learning method for identifying the best decision surface to maximise the separation between the different classes of fertilizers. In practice it performs quite well, particularly with classification on high-dimensional spaces – sometimes it is better than the standard baselines.

The decision function is given by:

$$f(x) = w^T x + b$$

where w is the weight vector and b is the bias term.

So the SVM classifier was kinda checked using both the full feature set and the Chi-square selected feature-set as well, just to be safe, you know.

D.1 Support Vector Machine without Chi-Square Feature Selection

So, the overall classification accuracy for the baseline SVM classifier was about 72% and the weighted F1-score was 0.68. Well it helped the classes that were dominant, but then it got weak at the knees for some of the minority classes – mostly due to features that were redundant and, also, to the class imbalance.

Table VIII. Classification Report of SVM (Without Chi-Square)

Class	Precisi on	Reca ll	F1-Sco re	Suppo rt
Compost	0.46	0.49	0.47	199
DAP	0.76	0.90	0.82	584
MOP	0.71	0.68	0.70	282
NPK	0.00	0.00	0.00	128
SSP	0.00	0.00	0.00	37
Urea	0.80	0.94	0.87	620
Zinc Sulphate	0.39	0.25	0.31	150
Accuracy	–	–	0.72	2000
Macro Avg	0.45	0.47	0.45	2000
Weighted Avg	0.65	0.72	0.68	2000

D.2 Support Vector Machine with Chi-Square Feature Selection

Changing the SVM classifier to use the Chi-Square feature selection improved the accuracy of the general classification from 72% to 83% and the weighted F1-score from 0.68 to 0.82. So that selected feature slice, the classes were more clearly defined plus noisier overlapping attributes were less impactful leading to more accurately identified fertilizer.

Table IX. Classification Report of SVM (With Chi-Square)

Class	Precisio n	Recal l	F1-Scor e	Support
Compost	0.68	0.75	0.71	199
DAP	0.88	0.90	0.89	584
MOP	0.87	0.80	0.84	282
NPK	0.70	0.59	0.64	128
SSP	0.00	0.00	0.00	37
Urea	0.91	0.92	0.91	620
Zinc Sulphate	0.59	0.71	0.64	150
Accuracy	–	–	0.83	2000

Macro Avg	0.66	0.67	0.66	2000
Weighted Avg	0.82	0.83	0.82	2000

V. ERROR METRICS

They have been used to assess the performance of machine learning models in the fertilizer recommendation system, and yes, it's just to ensure that the results are reliable. In this project you will make some models such as Random Forest, Gradient Boosting, XGBoost and Support Vector Machine (SVM) and you will evaluate them using various classification metrics, and your results are accurate, stable and reliable.

1. Accuracy

The focus of the evaluation for this project is accuracy. The ratio of the number of correct fertilizer classes to the number of all classes predicted. So it gives a rough overview of whether the model is performing well or not, but it may not suffice if there is an imbalance in the data, as the majority class might bias the outcome and the minority class may be overlooked.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$$

Where:

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives

2. Precision

The closer to the actual fertilizer categories the prediction is to the right the greater the precision. It is essentially the proportion of hits from all of the items the model classifies as “this is fertilizer X.” It's sort of like comparing the number of times the model got the fertilizer right, of all the times it had guessed at what fertilizer it was. The high precision also means there will be fewer false positives, which means the model will be less likely to identify a certain fertilizer when it shouldn't.

$$Precision = \frac{TP}{TP+FP}$$

The higher the precision value, the less likely the model will be to make incorrect fertilizer recommendations, and in some sense the more precise it is in its original recommendations.

3. Recall

The number of the real fertiliser cases that are correctly identified by the model is indicated by recall sort. Right, if there is high recall then it means that there are fewer false negatives, in this case the model missed fewer things that it should have picked up.

$$Recall = \frac{TP}{TP+FN}$$

With a higher recall value, the model can recall most of the correct fertilizer recommendations, you see. It sort of means the model gets a lot of the right ideas as well, not just the obvious ones.

4. F1-Score

The F1 score, a harmonic average of precision and recall, is kind of a balanced measure of the model's performance. When the data set is somewhat skewed, it's really helpful as pure accuracy can be misleading. And yes, because it's done using precision and recall, the F1-score also takes into account the false alarms or missed cases, false positives and false negatives. This is sort of a nice middle ground on that spectrum without having to look at just one metric.

$$F1 - Score = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

The higher the F1-score, such as, the better the trade off between the precision and recall. It's the same concept, but it's a balanced one you know?

5. Confusion Matrix

A confusion matrix is a tabular representation—a bit visual and very detailed—used to evaluate the performance of a classification model. It displays the number of True Positives (TP), True Negative (TN), False Positive (FP), and False Negative (FN) which can help us to better visualize how well the model is guessing/predicting. In this format, you get a good idea of what's going right, and what's going wrong – although some cases may appear messy in the beginning.

Actual / Predicted	Positive	Negative
Positive	TP	FN
Negative	FP	TN

Through a closer inspection of these values, it can be inferred what types of mistakes that the model made, and then one can also calculate what is important such as accuracy, precision, recall, and even F1-score.

6. Log Loss

Log Loss (also known as Logarithmic Loss) is a measure for assessing the performance of a classification model, but it does it by testing the predictions of the probabilities, not just the final predicted class. It tends to penalize incorrect answers that fail to be sufficiently tentative. It is not surprising that it is more stringent than simply measuring accuracy even when the model is near.

$$Log Loss = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^M y_{ij} \log(p_{ij})$$

Where:

- N = Number of samples
- M = Number of classes
- y_{ij} = Actual class label
- p_{ij} = Predicted probability

Lower log loss indicates that the model is performing better, with better predictions and a bit more confidence in those predictions; that's, the behavior is better.

TABLE X: Comparative Performance of Machine Learning Models Before and After Chi-Square Feature Selection

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	0.88	0.88	0.88	0.88
Gradient Boosting	0.87	0.88	0.87	0.87
K-Nearest Neighbors (KNN)	0.80	0.79	0.80	0.79

Support Vector Machine (SVM)	0.83	0.82	0.83	0.82
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It is pretty obvious from Table V that the comparative performance of the proposed classifiers before and after Chi-Square feature selection. There is small improvement in performance of KNN and SVM upon feature selection, with Random Forest showing the least improvement and remaining the best option across the board. This is sort of a proof of that – removing irrelevant features, even if these feature seem like harmless ones, improves classification accuracy and the model feels better overall.

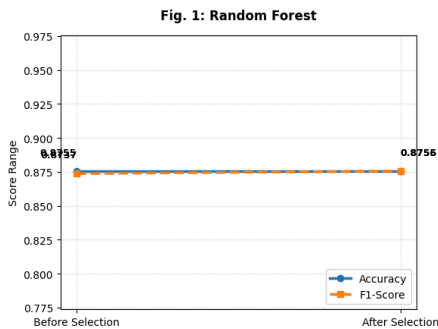
VI. PERFORMANCE ANALYSIS

The comparative analysis shows that ensemble learning models kind of beat the more traditional classification approaches, in fertilizer recommendation tasks . Overall performance appears to be better, although the base lines are okay some of the time.

A. Random Forest (RF)

The best performance overall was for the Random Forest classifier, while it was also relatively stable of the classifiers that were tested. With Chi-Square feature selection, this method got an accuracy of 88% and weighted F1-Score of 0.87 when this method was not used. Removing redundant or additional features did not negatively impact the accuracy (88%), but did improve the weighted F1 (0.88), so yea, it looks like they took the redundant and extra features out without it really hurting the prediction accuracy. Generally, these results show that Random Forest is reliable on complex agricultural information and maintains classification performance without any changes in the feature set.

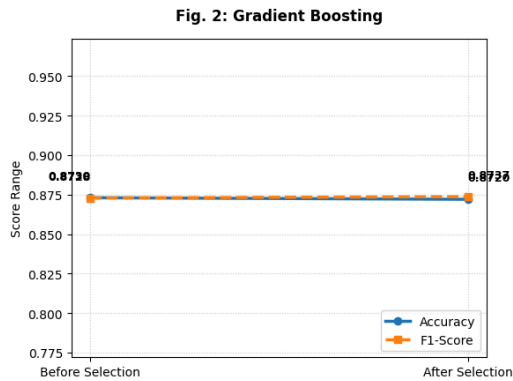
Fig. 1. Performance Comparison of Random Forest Before and After Chi-Square Feature Selection.



B. Gradient Boosting (GB)

Gradient Boosting also sort of produced competitive classification performance for the fertilizer recommendation, which is sort of good. The baseline model came in with an overall accuracy around 87% , and after applying Chi-Square feature selection it basically kept that same overall accuracy but with a small uplift in the macro-average metrics . This suggests that Gradient Boosting is effectively making use of the selected features , while still preserving its predictive ability, even if nothing much changes overall.

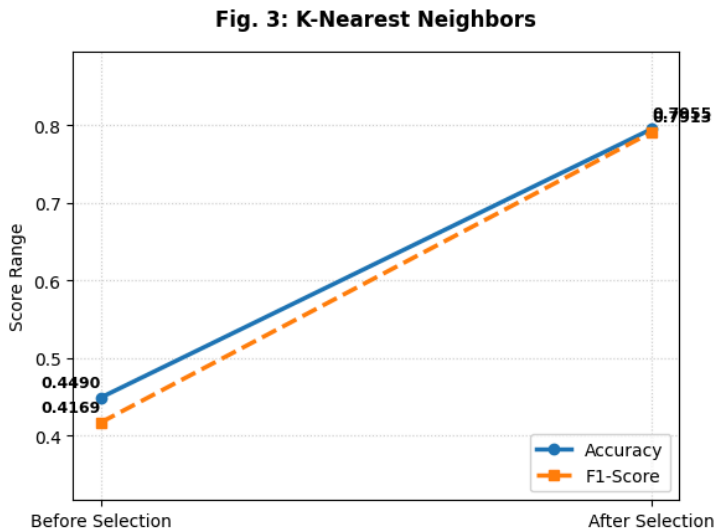
Fig. 2. Performance Comparison of Gradient Boosting Before and After Chi-Square Feature Selection.



C. K-Nearest Neighbors (KNN)

After feature selection, the K-Nearest Neighbors classifier had the most significant lift, meaning you know it really depends. The Chi-Square based approach had much higher accuracy levels up to 80%, whereas the baseline run had an overall accuracy of 45%. These features were irrelevant and when they were removed the dimensionality was reduced and the overall feel of this distance-based classification was sharper, and so all the measures of Precision, Recall and the F1-Score rose significantly. In general, this indicates that KNN is highly dependent on the process of feature selection.

Fig. 3. Performance Comparison of K-Nearest Neighbors Before and After Chi-Square Feature Selection.

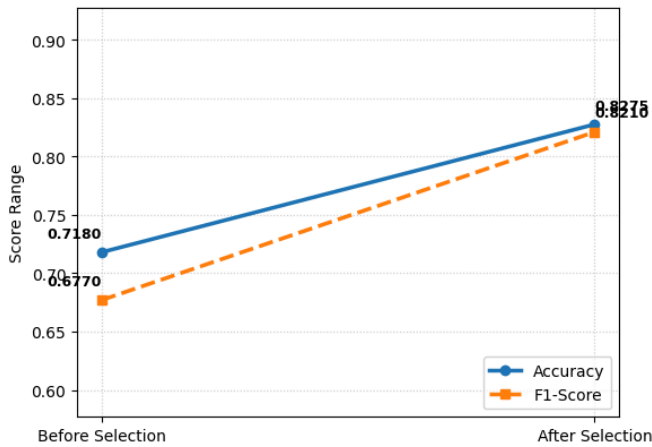


D. Support Vector Machine (SVM)

Gradient Boosting did somewhat contribute to the classification performance that was competitive for the fertilizer recommendation task, which is kind of good. The overall accuracy of the baseline model was approximately 87% and after using Chi-Square feature selection, the overall accuracy was approximately the same but with slightly improved macro-average scores. This implies that Gradient Boosting is indeed using the chosen features, although there is no significant change overall.

Fig. 4. Performance Comparison of Support Vector Machine Before and After Chi-Square Feature Selection.

Fig. 4: Support Vector Machine



VII. DISCUSSION

The experimental analysis illustrates why machine learning techniques for fertilizer recommendation can be beneficial, and it exemplifies the magnitude of change in model performance when using the feature selection method of Chi-Square. Four common machine learning classifiers, namely Random Forest (RF), Gradient Boosting (GB), K-Nearest Neighbors (KNN) and Support Vector Machine (SVM) were evaluated using various performance metrics such as Accuracy, Precision, Recall and F1-Score. Although the original data set appears to be “full”, it is evident from the comparison that we should be able to remove irrelevant redundant input attributes from the data by use of feature selection.

The overall classification accuracy of the considered model with highest and most stable performance was Random Forest which achieved accuracy of 88%. Applying Chi-Square feature selection kept it rather consistent, indicating it was robust as the agricultural data become more complicated or messy. Next was Gradient Boosting with an accuracy of 87%, and fairly balanced Precision (86%), Recall (86%), and F1-Score (86%). As a whole, it can be concluded that the ensemble learning method has potential in understanding the root causes relating the soil properties to recommendations that consistently yield a quality prediction.

In contrast, KNN and SVM seemed to show some considerable improvements, after applying Chi-Square feature selection. Indeed, a decrease in the dimensionality of the features showed an increase in the accuracy of KNN from 45% to 80% and for SVM from 72% to 83% which indicates that the classifiers which are sensitive to the irrelevant features improved at the reduction in the dimensionality of the features. The results of the experiments indicate that the Chi-Square feature selection improves the efficiency and performance of the proposed fertilizer recommendation system, while retaining the strong classification ability of the machine learning methods used.

VIII. CONCLUSION AND FUTURE WORK

In this study, four machine learning algorithms Random Forest, Gradient Boosting, K-Nearest Neighbors (KNN) and Support Vector Machine (SVM) was implemented with the objective of fertilizer recommendation, which treated the problem as a sort of comparative analysis with the use of Chi-Square feature selection. The developed system was evaluated using the accuracy, precision, recall and f1 score to determine the practical efficient of each one of the classifiers. Based on the experiments, it appeared that the accuracy of classification obtained by Random Forest was at approximately 88% and Gradient Boosting came next at approximately 87%. Also, the Chi-Square feature selection has definitely helped boost the results for KNN and SVM and it does sort of verify that effective feature selection, especially

picking features of consequence/relevance to the problem does boost the predictive power of the machine learning problem.

The recommended system of fertilizers provides a more effective route for making fertilizer recommendations, when "soil plus environmental parameter" data are utilized. The system improves classification accuracy by taking out redundant functions before training the model and reduces the complexity of a little bit. The results demonstrate that the feature selection is one of the most important steps in agriculture decision-making in the context of increasing the accuracy and reliability of machine learning models.

With this, further steps will be taken based on this comparative study and more feature selection methods will be tested as well, such as Mutual Information and Random Forest Feature Importance, just for the fun of seeing what's going on in another view. The entire proposed system could also potentially be evaluated within larger and more diverse sets of agricultural data, preferably across several geographic sectors, to render a less limited result. Additionally, the integration of the recommendation model within a web-based or mobile application and linking it with IoT sensors and soil monitoring can bring up the possibility of providing real-time fertilizer suggestions and enhance the efficacy of precision farming practices.

